

Flexible pooled phenotyping enables population-scale mapping of RNAi-sensitivity modifiers

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Abstract

Genetic interactions contribute to phenotypic variation, yet most studies characterize them in a small number of reference backgrounds rather than across natural populations. Studying genetic interactions in *Caenorhabditis elegans* is facilitated by RNAi by feeding, which enables gene-specific knockdowns across many genetic backgrounds without the need to construct individual mutant lines. However, individually screening hundreds of isolates across many RNAi conditions is laborious. Here, we developed a pooled phenotyping platform that uses whole-genome sequencing to quantify the fitness of hundreds of genetically distinct *C. elegans* strains competing in a single flask. We leveraged this platform to evaluate RNAi responses in a pooled population of RNAi-sensitive *C. elegans* strains and identified two classes of QTL that influence RNAi responses. The first class influences responses to knockdowns of specific genes and is exemplified by a QTL that is specific to knockdown of *mig-6*. The second class influences general RNAi-response across multiple knockdowns. One QTL in this class localized to the right arm of chromosome III and was detected by both GWA and xQTL mapping, which suggests that a common large-effect variant underlies this QTL. We identified a T96K missense change in the dsRNA receptor SID-2 as the underlying causal variant. The 96K allele resides in the extracellular domain of SID-2 and confers increased RNAi sensitivity relative to the 96T allele. We hypothesize that the 96K allele enables greater uptake of ingested lumenal dsRNA through higher-affinity binding to the dsRNA phosphate backbone. The platform we developed enables systematic mapping of genetic interactions across natural populations at a previously impractical scale.

Introduction

The same genetic lesion does not produce the same outcome in every individual who carries it. This shows up as incomplete penetrance and variable expressivity in Mendelian disease, where carriers of an identical pathogenic variant range from unaffected to severely affected. A growing body of work indicates that rare and common background variation is a major source of this variation (Heyne et al., 2026; Kang & Drivas, 2026). For example, Huntington's disease risk is conferred entirely by an *HTT* CAG repeat expansion, but background variation with two independent effects on chromosome 15 moves onset 6.1 years earlier and 1.4 years later, and a chromosome 8 locus hastens onset by 1.6 years (Genetic Modifiers of Huntington's Disease (GeM-HD) Consortium, 2015). Biobank-scale sequencing is beginning to identify rare large-effect modifiers, but such modifiers remain difficult to detect because isolating the effect of one variant against a fixed, inherited background requires exceedingly large samples (Kang & Drivas, 2026). These observational studies establish that genetic background is a major determinant of disease phenotypes, but they remain observational because neither the causal mutation nor background can be chosen or resampled.

Model organisms don't have these constraints because a single perturbation can be introduced into many natural genetic backgrounds and its effect measured directly. Modifier screens at scale show that dozens of essential genes in the laboratory reference yeast strain are dispensable in at least one natural genetic background (Batté et al., 2026; Parts et al., 2021). These studies also reveal that the genetic modifiers of knockout mutations tend to map to single modifier genes. However, extending these studies to metazoan organisms is largely intractable because it requires constructing hundreds of individual mutant lines. *Caenorhabditis elegans* doesn't have this limitation because RNAi by feeding enables knock down of target genes without altering the genome (Timmons et al., 2001; Timmons & Fire, 1998). Despite the unusual sensitivity of *C. elegans* to ingested RNAi (Nuez & Félix, 2012), individually measuring hundreds of strains across many knockdown conditions has limited the breadth of this approach. The most extensive such screen assayed 29 maternal-effect genes across 55 wild strains and recovered highly polygenic architecture driven by gene-specific modifiers (Paaby et al., 2015).

Pooled selection and sequencing can enable simultaneous phenotyping of many genotypes in a single flask. In *C. elegans*, this has been implemented using molecular inversion probes (MIP-seq) to track strain identity within a pooled competition. This approach was implemented to study variation in competitive fitness of 22 *C. elegans* isolates in the presence of microsporidia (Mok et al., 2023), and resolved a starvation-resistance locus identified by pooled phenotyping of 100 genetically diverse strains (Webster et al., 2022). A major limitation to this approach is the construction of population-specific probe panels before phenotyping can begin, which can limit panel size and ties reagents to one fixed set of strains. By contrast, inferring strain frequencies directly from whole-genome sequencing only requires sequenced genotypes that are publicly available on CaenDR (Cook et al., 2017; Crombie et al., 2024).

Extending pooled selection experiments to identify genetic modifiers of a specific knockdown must contend with heritable variation in RNAi sensitivity across the *C. elegans* population (Chou

et al., 2024; Elvin et al., 2011; Paaby et al., 2015; Pollard & Rockman, 2013). A panel of wild isolates therefore segregates general RNAi modifiers that shift responses of every knockdown and knockdown-specific modifiers. We therefore designed an experiment that could simultaneously map general RNAi-sensitivity variation and identify potential genetic modifiers of knockdown perturbations.

Here we present a pooled phenotyping platform for *C. elegans* that infers strain frequencies from bulk sequencing using non-negative least squares regression and show that it performs comparably to a previously published MIP-seq experiment (Webster et al., 2022). We used this platform to phenotype 231 wild *C. elegans* isotypes exposed to double-stranded RNA targeting *pos-1* and identified multiple QTL that explain RNAi-response variation across the population. Of the identified QTL, one that localizes to the right arm of chromosome III was independently identified in an xQTL experiment (Burga et al., 2019; Zdraljevic et al., 2025). We fine-mapped this region to a single residue substitution, T96K, in the extracellular domain of the dsRNA receptor SID-2. We show that this variant increases RNAi sensitivity in multiple genetic backgrounds, including the standard laboratory strain, N2. The T96K shifts responses to multiple knockdowns because it likely affects RNAi delivery to the intestine. Contrasting two knockdown conditions isolates the target-specific component and reveals multiple large-effect loci modifying the response to a single gene knockdown.

Results

Accurate strain frequency inference from whole-genome sequencing

If a pool of wild isolates is grown under selection and sequenced in bulk, can per-strain frequencies be inferred from alternate-allele counts by non-negative least squares regression against a CaeNDR genotype matrix (Methods). To answer this question, we used simulations to seed seven independent populations using published *C. elegans* traits as fitness (variable pool sizes ranging from 86 to 330, based on trait; Methods). We simulated observed alt-allele counts across a range of sequencing depths (1-500x), which were deconvolved back to per-strain frequencies using NNLS regression. Comparing the estimated frequencies to the input revealed that NNLS recovered strain frequencies with a median r^2 of 0.906 at 10X depth (IQR: 0.786-0.957) and rising above 0.986 at 30X depth and above (**Figure S1**)

We next tested the same inference on pooled wild isolates of known compositions. We extracted DNA from populations composed of 46 and 40 strains, combined the extracted DNA at known ratios, and inferred individual strain frequencies. Because we did not pool the strains at known frequencies, we summed the strain frequencies derived from each pool to establish an input pooled population ratio. We found that the inferred shares tracked the designed ratios closely (Pearson $r = 0.997$, root mean squared error 0.038 in fraction units), with the largest deviations occurring at the smallest-volume step of the dilution (**Figure S2**).

As a final test for the NNLS approach, we compared it with a previously published pooled-phenotyping experiment that used molecular inversion probe sequencing to infer strain frequencies (Webster et al., 2022). To make the comparison, we performed whole-genome sequencing on the same samples. For each strain, we took the rate of change in pool frequency across days of L1 starvation as the slope of frequency change regressed on day and averaged those slopes over the five replicate arms (Methods). We applied this analysis to both platforms' frequency estimates to compare measurements and not analysis choices. We found the NNLS-derived slopes to be highly correlated with those derived from MIP-seq (Spearman's $\rho = 0.974$, $n = 98$ strains, $p < 1e-4$) (**Figure 1A**). Agreement was largely retained as we downsampled reads, reaching $\rho = 0.85$ at 1x (**Figure S3**). When we compare the NNLS-derived slopes to the published *Slope* trait, which is fitted to $\log_2$ -transformed frequencies, we observe a correlation of $\rho = 0.89$ at full depth and 0.74 at 10x depth. Taken together, these three validation experiments gave us confidence that we could infer per-strain frequency dynamics accurately enough to treat them as a quantitative phenotype.

Pooled phenotyping enables evaluation of wild isolate RNAi responses

C. elegans RNAi screens are routinely performed in the N2 strain. However, extending these screens to multiple wild isolates is challenging because each isolate needs to get assayed independently. We reasoned that the pooled phenotyping approach would facilitate fast evaluation of wild isolate RNAi responses. N2 strains exposed to *pos-1* RNAi exhibit an embryonic lethal phenotype, making this treatment amenable to the pooled phenotyping because RNAi-responsive strains will fail to contribute progeny to subsequent generations. To explore if we could quickly evaluate wild isolate RNAi responses, we performed a pilot experiment where we exposed 231 pooled wild isolates to *pos-1* RNAi-expressing HT115 bacteria or control HT115 bacteria (Methods). We grew these populations for two generations in these conditions across four *pos-1* replicate pools and two control pools, sequenced the resulting F3 L1 populations, and inferred the individual strain frequencies in each population. We calculated individual strain frequency differences between the control and *pos-1* RNAi conditions and found good agreement between the replicate conditions (**Figure S4**; Spearman's $\rho = 0.77-0.87$ across all six pairwise replicate comparisons). 183 of the 231 wild isolates (79%) declined in frequency in *pos-1* RNAi relative to the control condition, a significant excess over the even split expected if frequency changes reflected noise alone (binomial test, $p < 2 \times 10^{-16}$) (**Figure 1B**). Because pool frequencies are compositional, this proportion measures decline relative to the pool, rather than absolute per-strain responses. In total, 81 isolates are absent from every *pos-1* pool while present in both controls. Of the remaining 150 individuals whose frequencies vary across *pos-1* replicates, 60 decline in frequency (one sample t-test on four replicate deltas $q < 0.05$). Together 141 of 231 isolates (61%) are individually responsive to *pos-1* RNAi. We performed a genome-wide association scan on the *pos-1* responses, which identified QTL on the right arm of chromosome IV and the left arm of chromosome X (X:4,875,969) that passed the Bonferroni significance threshold and a third QTL on the right arm of chromosome III above a less conservative threshold (Methods) (**Figure 1C**). From this analysis, it is unclear if these QTL are *pos-1*-specific or general RNAi-response QTL. We reasoned that if they are general RNAi-response QTL, we would want to minimize their contribution to future experiments. Therefore, we constructed a pooled population of RNAi-responsive strains. To construct this population, we manually re-evaluated *pos-1* RNAi responses of 191 wild strains on agar plates and identified 93 strains with robust RNAi responses that we pooled to evaluate additional RNAi responses (Methods). We compared the manual *pos-1* RNAi phenotypes we collected to the pooled RNAi responses based on strain frequency changes above and found agreement between the methods (Spearman's $\rho = 0.41$, $p = 7.8 \times 10^{-6}$; $n = 111$, **Figure S5A**). We also found good agreement between our manual phenotypes and a previously published evaluation of wild isolate RNAi responses (Spearman's $\rho = 0.55$, $p = 0.014$; $n = 19$; **Figure S5B**) (Paaby et al., 2015).

Knockdown-specific and general RNAi-response QTL segregate in wild isolates

Having assembled a pool of RNAi-responsive strains, we next asked whether we could identify QTL that modify responses to individual RNAi treatments. We grew this pool on bacteria expressing double-stranded RNA against each of ten target genes and on the HT115 empty-vector control, sequenced the resulting populations, and inferred strain frequencies (Methods). We performed genome-wide association scans on the responses to these knockdowns and identified QTL that exceeded the Bonferroni threshold for *pos-1* and *mig-6* RNAi (Figure 2). The *pos-1* response mapped to the right arm of chromosome III (III:12,353,680, $-\log_{10}p = 7.04$, allele frequency 0.11), just 0.4 Mb from the association we identified in our pilot experiment (Figure 1C). The *mig-6* response mapped to the center of chromosome V (V:14,647,434, $-\log_{10}p = 7.71$, allele frequency 0.13), a locus we did not identify in either *pos-1* experiment, suggesting this locus is *mig-6*-specific.

A pooled scan can only detect common alleles in the panel, and because each strain's phenotype is an inferred frequency, detected QTL could in principle reflect inference error and not a response to knockdown. A cross addresses both of these limitations because parental alleles segregate at equal frequencies and those frequencies are directly measured. We therefore exposed two crosses to *mig-6* and *pos-1* RNAi after intercrossing for ten generations (Methods). One cross was constructed between JU1793, which was resistant to both *mig-6* and *pos-1* RNAi, and JU2466, which was sensitive to *pos-1* RNAi and had an average *mig-6* response (Figure S6). We made use of a previously constructed cross between N2 and XZ1516 because XZ1516 had among the strongest responses to both RNAi conditions (Figure S6) (Zdraljevic et al., 2025). We sequenced the cross populations after two generations on the RNAi conditions and used allele frequency deviations between conditions to perform QTL mapping. By comparing parental frequencies after RNAi exposure to the control HT115 condition, we identified multiple QTL shared between the crosses and cross-dependent QTL (Figure S7B-C, Supplemental table X). Additionally, we identified QTL that are shared between conditions, which we refer to as general RNAi-response QTL. By contrasting allele frequencies after *pos-1* and *mig-6* RNAi exposure, we identified RNAi-specific QTL. One major general RNAi-response QTL we identified in both crosses after *mig-6* and *pos-1* exposure localized to the right arm of chromosome III (Figure S7B-C), in close proximity to the QTL we observed from the pooled association mapping (Figure 2). We also identified a large *mig-6*-specific QTL in the XZ1516 cross that co-localized with the *mig-6*-specific QTL we identified in the pooled experiment. In addition to these re-discovered QTL, we identified two novel large-effect *mig-6*-specific QTL on chromosome I in the N2 x XZ1516 cross, where parental allele frequency differs by 0.391 (95% CI 0.378-0.404) between the *mig-6* and *pos-1* conditions and one on chromosome X in the JU1793 x JU2466 cross, differing by 0.713 (95% CI 0.679-0.747). Both of these effects are essentially unchanged when contrasted to the HT115 empty-vector control (0.386, 95% CI 0.373-0.399 and 0.803, 95% CI 0.775-0.831), indicating they track responses to *mig-6* knockdown and not general RNAi responses. Independently recovering the chromosome III and V loci by crosses confirms they are not artifacts of deconvolution. Taken together, these results highlight the power of the pooled-phenotyping approach to measure RNAi responses across the

C. elegans population, point to substantial RNAi-response variation segregating in the population, and map several loci that modify gene-specific knockdown responses.

A high-frequency variant in the dsRNA transporter SID-2 confers increased sensitivity to ingested RNAi

We next sought to uncover the variant underlying the general RNAi-response QTL on the right arm of chromosome III. To make near-isogenic lines (NILs), we first confirmed that JU2466 (35.7% hatched; 95% CI 29.7-42.0%) and JU1793 (99.4% hatched; 95% CI 96.9-99.9%) had distinct *pos-1* RNAi phenotypes using a plate-based embryonic lethality assay (**Figure S7**; Methods). Next, we introgressed the chromosome III QTL region from JU2466 into the JU1793 genetic background and found it conferred *pos-1* sensitivity to JU1793 and was not significantly different from JU2466 (39% hatched; 95% CI 32.8-45.6%). Through iterative NIL construction we localized the QTL to a 37 kb interval spanning 13.658–13.695 Mb (**Figure 3C**; **Figure S7**). Within this region we found 27 parental differences across twelve genes, of which two alter a protein and both are missense mutations in *sid-2* (V5L and T96K). *sid-2* encodes a single-pass transmembrane protein localized to the apical membrane of intestinal cells that facilitates uptake of dsRNA from the gut lumen, making it an excellent candidate to harbor variation that contributes to RNAi sensitivity across wild isolates. We focused on the T96K variant for two reasons. First it is also the allele that distinguishes the parents of the N2 x XZ1516 cross, in which an overlapping general RNAi-response QTL is segregating that we assume might be driven by the same underlying variant (**Figure S8B-C**). Second, the residue 96 is predicted to reside in the extracellular domain of SID-2 and might alter interactions with double-stranded RNA.

We generated reciprocal allele-swap strains to test whether this variant was contributing to *pos-1* RNAi responses. Introducing the 96K allele into the JU1793 strain lowered the strain's hatching rate from 94.8% (95% CI 90.9-97.4) to 53.1% (95% CI 46.1-60.1), while introducing the 96T allele into the JU2466 strain raised its hatching rate from 4.5% (95% CI 2.8-7.0) to 18.4% (95% CI 13.5-24.2) (**Figure 4a**). Therefore, the 96T reference allele produces a weaker RNAi response than the alternate 96K allele. These results suggest that this allele interacts with other variants in the parental strain's background to produce background-dependent effects: the 96T allele raises hatching odds 16.1-fold in JU1793 (95% CI 8.2-34.8) but only 4.0-fold in JU2466 (95% CI 1.9-8.8), a 4.08-fold difference between backgrounds (95% CI 1.56-10.71, likelihood ratio $p = 0.0049$)

Because N2 is the standard laboratory reference strain and has been used in countless successful RNAi experiments, we were surprised to find that the 96T reference allele caused a weaker RNAi response. We therefore asked if we could increase the RNAi sensitivity in N2 by introducing the 96K allele. Under our standard assay condition (50:50 mixture of *pos-1* and HT115 bacteria) introducing 96K into the N2 strain produced no difference in the embryo hatching rate. This is likely because at this dose N2 fails to hatch 94% of its embryos (6% hatched, 95% CI 3.2-10.0) and there is little room for further sensitization (**Figure S9**). However,

when we lowered the *pos-1* bacteria concentration on the assay plates, we found that the 96K allele lowered the N2 hatching rate from 32.3% (95% CI 26.4-38.7) to 4.0% (95% CI 2.6-6.0) (two independently edited lines, 4.4% and 3.7%; **Figure 4B**), therefore increasing RNAi sensitivity in this background as well.

96T is the third position of a predicted N-linked glycosylation sequon (consensus N-x-S/T), N94-C95-T96. The T96K allele destroys the sequon and adds a positive charge. Because N-linked glycosylation of membrane proteins typically supports folding, stability, and membrane localization, and because this sequon is conserved across nine of ten species of the *Elegans* supergroup and its sister *C. kamaaina* (**Figure S10A**), loss of the site would be predicted to impair SID-2 function. Although there is no experimental evidence of N94 being glycosylated in *C. elegans*, we observe the opposite, with our result that the 96k allele is more sensitive to ingested double stranded RNA than T96.

To separate the loss of the sequon from the introduction of the lysine, we introduced a N94A allele into both JU1793 and JU2466. The N94A allele caused a decrease in *pos-1* RNAi responses in both genetic backgrounds, consistent with disrupted SID-2 function (**Figure S10B**). Because both alleles break the sequon in JU1793, these results indicate that the charge introduced by the lysine reverses the effect of losing the glycosylation sequon. Since the sequon is already disrupted by the 96K allele in JU2466, this result suggests the asparagine contributes to SID-2 function independently of an attached glycan. Consistent with this, the N94A in the JU2466 background conferred more resistance than restoring the ancestral residue at position 96 (42.5% versus 18.4%, OR 3.3, $p = 9e-8$). This is difficult to reconcile with the two residues acting through a shared modification.

We hypothesized that by introducing a positively charged residue the T96K allele might facilitate a stronger interaction with the phosphate backbone of double-stranded RNA and therefore better uptake of the double-stranded RNA. To assess this, we computed the local net charge in 12 Å windows across the SID-2 ectodomain. The window centered on residue 96 has a net charge of +1.24e. This is the 82 percentile of all ectodomain residues and is largely driven by K93 and K132 sitting at 6.6 and 6.8 Å away. (**Figure 4C**, **Figure S11**). Introducing the 96K variant to this local region shifts the net charge to +2.24 e, which corresponds to the 98th percentile of the ectodomain's local net charge distribution (**Figure S11**). While these observations don't prove that the T96K allele is causing a stronger interaction with double-stranded RNA by increasing the net charge of its local environment, it does agree with previous reports that modifying critical histidines in the ectodomain to positive arginines is less disruptive to uptake of double-stranded RNA than modifying the histidines to alanines (McEwan et al., 2012).

Discussion

We developed a phenotyping platform that can reliably infer the frequencies of hundreds of strains in pooled populations. This platform can readily be applied to any pooled population of strains with known genotypes because it relies on whole-genome sequencing. Demonstrating this point, we first used this platform to map RNAi-response variation across 231 *C. elegans* isolates to three distinct QTL (**Figure 1B-C**). We then applied it to an 84-isolate pool of RNAi-sensitive individuals that identified the same general RNAi-response QTL on chromosome III and an additional knockdown-specific QTL on chromosome V (**Figure 2**). We validated these QTL using two independent bi-parental crosses that revealed several other modifiers of *mig-6* and *pos-1* RNAi (**Figure S8B-C**).

We resolved the general RNAi-sensitivity locus on the right arm of chromosome III to a T96K missense mutation in the extracellular domain of the dsRNA receptor SID-2. We show that strains with the 96K allele are more sensitive to *pos-1* RNAi than strains with the 96T allele across three genetic backgrounds, including the standard laboratory strain N2 (**Figure 4A-B**). T96 sits in a conserved N-glycosylation sequon, N94-C95-T96, and while the T96K allele adds a positive charge to the ectodomain that might facilitate stronger interaction with the phosphate backbone of double-stranded RNA, it should also disrupt the conserved sequon. Because N-glycosylation is associated with protein stability and localization, disrupting the sequon should impact SID-2 function. However, we see the opposite effect in the T96K allele swap experiments. We therefore introduced an N94A allele in both genetic backgrounds to disrupt the sequon independently of T96. The N94A allele makes JU1793 and JU2466 more resistant to *pos-1* RNAi, suggesting that this allele impairs SID-2 function (**Figure S10**). Because T96K also disrupts the sequon, while contributing a positive charge, we suspect that the electrostatic effect of T96K must be large enough to override the antagonistic, glycosylation-linked resistance effect we observed in the N94A edited strains. We hypothesize that by adding localized positive charge to the ectodomain, the lysine allele facilitates more double-stranded RNA uptake than the threonine thereby inducing a stronger RNAi response in carrier worms. We note that we provide no experimental evidence that the N94-C95-T96 N-glycosylation sequon is glycosylated and there is no published experimental evidence suggesting it is (Hu et al., 2026). The absence of detected SID-2 glycosylation in this dataset is likely driven by sampling depth in the proteomics dataset because only 870 proteins were recovered (Hu et al., 2026). In proteomes that do detect SID-2, only the cytoplasmic domain is detected, not the N94-C95-T96 peptide (Narayan et al., 2016; Tan et al., 2024).

In addition to the identification of general RNAi-response QTL, pooled RNAi phenotyping and a bi-parental cross revealed a *mig-6*-specific QTL localized to the center of chromosome V and the left arm of chromosome I (**Figure 2, Figure S8**). We hypothesize that these QTL are *mig-6*-specific and not a general RNAi-response locus because both N2 and XZ1516 are strong responders to RNAi, making disrupted RNA transport to *mig-6* expressing somatic tissues a less likely explanation. The chromosome V QTL is likely a common variant of large effect because it was identified by the pooled assay, while the variant underlying the chromosome I QTL is likely less frequent across the mapping population. Establishing these QTL as *mig-6* RNAi-specific

would require identifying the underlying variant or performing additional RNAi-knockdown experiments in this cross to verify the lack of a chromosome I and V signal in unrelated RNAi conditions. It is more likely that the apparent *mig-6*-specific QTL we identified on chromosome X in the JU1793 x JU2466 cross (**Figure S8**) is being driven by disrupted RNA trafficking to the target tissue because JU1793 has been shown to have inefficient somatic RNAi (Nuez & Félix, 2012).

These results highlight the platform's ability to identify large-effect genetic modifiers of genetic perturbations. Thousands of rare Mendelian are caused by loss-of-function variants and their penetrance and expressivity are frequently modified by background variation (Kang & Drivas, 2026). Such modifiers are currently difficult to identify in human genetics studies because rare disease-causing variants are carried by too few individuals to give modifier scans sufficient statistical power. The platform we developed enables us to apply a wide-range of genetic perturbations of disease-causing homologs to a pool of genetically diverse *C. elegans* isolates to identify genetic modifiers.

Methods

Strain Maintenance

Prior to plating on RNAi conditions, all *C. elegans* strains were maintained at 20°C on modified nematode growth medium (NGMA), containing 1% agar and 0.7% agarose and seeded with OP50. When needed, strains were shifted to 15°C for synchronization and developmental timing.

RNAi Bacteria

All RNAi bacteria were streaked onto LB + 100 µg/ml carbenicillin agar plates to form single colonies. Single colonies were grown overnight in 3 ml liquid LB + 100 µg/ml carbenicillin cultures. 24 hours later, the cultures were diluted 1:1000 in LB + 100 µg/ml carbenicillin and grown for 16 hours. Finally, RNA expression was induced with 1 mM IPTG for four hours. 100 µl of 1000 µl of these cultures was used to seed 6 cm or 10 cm plates, respectively. RNAi plates were dried for two days at room temperature and then transferred to 4°C.

After observing inconsistent RNAi responses in known responding strains, large volumes of RNAi bacterial cultures were grown, OD-standardized, and stored at 4°C in aliquots to reduce batch variability. Three liters of each culture were transferred to centrifuge bottles and centrifuged at maximum speed for 10 minutes. The supernatant was discarded, and the remaining bacterial pellet weight was calculated by subtracting the empty bottle weight from the pellet-containing bottle weight. The bacterial pellets were resuspended with K-media at a ratio of 1 g : 4 mL to generate 10x RNAi bacterial stocks. For plating convenience, stocks were further diluted to 1x and frozen at -80°C in 5 mL aliquots. All subsequent NIL phenotyping has been plated on NGMA + carbenicillin plates with a 1:1 mixture of pos-1 and HT115 cultures normalized at an OD₆₀₀ of 12.9.

NNLS Simulations

Genotype data. Genotypes for *C. elegans* wild isolates were obtained from the CeNDR release (2021-01-21) in PLINK binary format and read into R with the BEDMatrix package. Variants on each of the six chromosomes (I–V, X) were loaded separately and combined into a single genotype matrix **G** of dimension m markers $\times$ n strains. Missing calls were set to the reference state, and genotypes were binarized (homozygous-alternate coded as 1, reference as 0), which is appropriate given that wild isolates are nearly fully homozygous. From this matrix we precomputed the strain-by-strain Gram matrix **G^TG** for use in the deconvolution step.

Ground-truth model. Each strain i was assigned a fitness value w_i drawn independently from a scaled inverse- χ^2 distribution with 12 degrees of freedom and scale parameter 1 (extraDistr). The expected alternate-allele count at each marker was computed as the fitness-weighted sum over strains carrying the alternate allele, $\mathbf{c} = \mathbf{G}\mathbf{w}$, and converted to an expected pooled allele

frequency $\mathbf{p} = \mathbf{G}\mathbf{w} / \sum_i w_i$. The corresponding normalized strain frequencies, $\mathbf{w} / \sum_i w_i$, were retained as the ground truth against which inferred frequencies were compared.

Simulated pooled sequencing. We simulated observed alternate-allele counts across a range of target sequencing depths (5, 10, 30, 50, 100, and 500×; assuming a ~120 Mb genome and 300 bp reads). For each depth d , the observed count at marker j was drawn as $y_{\square} \sim \text{Binomial}(d, p_{\square})$, yielding a marker $\times$ depth count matrix $\mathbf{Y}$.

Deconvolution. Observed counts were projected onto the strain space as $\mathbf{G}^T\mathbf{Y}$, and per-strain coefficients were estimated for each depth by non-negative least squares (RcppML::nnls), solving the normal-equations form $(\mathbf{G}^T\mathbf{G})\boldsymbol{\beta} = \mathbf{G}^T\mathbf{y}$ subject to $\boldsymbol{\beta} \geq 0$ —equivalently minimizing $\|\mathbf{G}\boldsymbol{\beta} - \mathbf{y}\|^2$ over non-negative $\boldsymbol{\beta}$. The resulting coefficient vectors were normalized to sum to one to give inferred strain frequencies, which were then compared to the known input frequencies to evaluate recovery accuracy as a function of sequencing depth.

Creation of pooled populations

We created pooled populations of approximately 48 strains each. To create these pooled populations, we recovered individual CeNDR strains on 6cm NGMA plates for three days and then bleached them on new 6cm plates. The following day, L1s were transferred to new 6cm plates and were allowed to starve. Once all strains from a given plate were starved (~4 days), approximately 1 cm chunks from the starved plates were transferred to three different 10cm plates and grown for three days. After three days of growth, the worms from all the plates were washed off with M9 and put in a 50ml conical tube. Worms were then bleached to harvest embryos, and the embryos were resuspended in K media and arrested overnight as L1s. L1s were titred and diluted to contain approximately 1 million worms/ml and frozen for long-term storage. In total, we prepared eight of these populations, which we labeled A-H.

Dilution experiment

To perform the dilution experiment, we recovered the B and C pooled populations from the freezer in 1 L flasks that contain 200ml of K media and 1X HB101. We allowed the populations to grow for 72 hours until they were gravid, at which point we gravity settled the worms in 50 ml conical tubes and bleached the population. We arrested the populations in K media overnight and allocated approximately 500,000 worms for a DNA preparation. We combined the two DNA preparations at nine known ratios (B1 fraction: BC1 = 0.09, BC2 = 0.17, BC3 = 0.29, BC4 = 0.44, BC5 = 0.62, BC6 = 0.76, BC7 = 0.86) and prepared libraries for whole-genome sequencing.

Inferring strain frequencies

Raw sequencing reads were aligned to the N2 genome (build: WS276) using *bwa mem* (Li & Durbin, 2009). The GATK *ASEReadCounter* tool was run on the resulting BAM files using the CaeNDR VCF (release: 20220216) (Cook et al., 2017). For a given experiment, a given variant

site was retained if it had a mean depth between 10× and 150× across all samples and no missing depth values. Next, we removed nonvariable sites in the experimental strain population and sites with fewer than 1,000 genotype calls in the *C. elegans* Natural Diversity Resource (CeNDR) VCF. To correct for reference bias in allele frequency estimation, variants with a minor allele frequency greater than 50% in the experimental population were recoded so that the more common allele was always treated as the reference, with corresponding allele counts flipped accordingly. Strain frequencies in each bulk sample were estimated by non-negative least squares (NNLS) using RcppML. The genotype matrix (markers × strains) and per-sample alternate allele counts were used to solve for strain composition by minimizing the residual between observed allele counts and those predicted by the linear combination of strain genotypes, with solutions constrained to be non-negative and normalized to sum to one. Estimated strain frequencies were output for all samples as a strains × samples matrix.

Pooled GWA

Initial *pos-1* RNAi experiment:

A pooled population derived from sub-populations B, C, E, F, and G was thawed from a frozen stock directly into 500 ml of K media and 30 ml of 10x HB101 bacteria. After 72 hr of recovery, the gravid population was harvested and bleached in 50 ml conical tubes. Embryos were resuspended in K media and allowed to hatch and synchronize as L1s overnight. The following day, the synchronized L1s were titrated and used to seed the experiment. We set up replicate liquid cultures for *pos-1* and HT115 control conditions in 250 ml baffled flasks. We put ~75,000 worms in 50 ml of K media and 7 ml of 10X HT115 or *pos-1* RNAi bacteria in each flask and incubated them at 20°C shaking at 220 RPM for 72 hrs. Next, we bleached each of the four cultures and arrested L1s as described above. We repeated the growth conditions for the second generation. To harvest F3 L1s for sequencing, we transferred the cultures to conical tubes and allowed the F2 adults to settle for 5min. Next, we took the supernatant containing L1s and embryos through 20 µm filters, which allowed the L1s to pass and retained the embryos. We spun down the separated L1s at 1200 RPM for 2 min, removed the liquid from the cultures and transferred the pellets to microcentrifuge tubes that were stored at -80 °C until sequencing. DNA extraction, library preparation, and sequencing are described in another section.

Construction of an RNAi-sensitive *C. elegans* population:

We used *pos-1* RNAi to evaluate *C. elegans* RNAi responses prior to pulling the strains. We thawed wild isolates that we purchased from CeNDR on 10 cm NGMA plates seeded with OP50 and incubated them at 20°C until gravid adults were present on the plate. We bleached ~15 adults from individual strains on 6 cm NGMA plates seeded with OP50. The following day we transferred approximately 10 L1 larvae to fresh 6 cm NGMA plates seeded with OP50 and incubated them at 20°C until L4s (F1) from the next generation were present. We then transferred 5 L4s to fresh NGMA agar plates seeded with OP50 and waited until the following generation were L4s (F2s). We transferred 10 L4 worms to NGMA + Carbomycin + IPTG plates that were seeded with induced *pos-1* RNAi bacteria. The following day we removed the adults

from those plates and allowed the embryos to hatch overnight. We defined a qualitative ranking of the hatching rate we observed across the wild isolates from 0 (no embryos hatched) to 5 (all embryos hatched) and assigned each wild isolate a score. Because there were not enough strains that had a complete RNAi response, we pooled strains with scores of 0 or 1, which corresponded to fewer than 5 embryos hatching to larvae. To pool the strains we went back to the plate we sourced the worms that we placed on *pos-1* RNAi, which was starved. For each strain we decided to put in the pooled population, we chunked from the starved plates to two 10 cm NGMA plates seeded with OP50 and allowed them to grow for approximately 96 hrs until they were almost starved. We washed all the strains off of the 10 cm plates into a 50 ml conical tube and allowed the adults to settle. We aspirated the supernatant and bleached the population to harvest embryos. The following day we titered the arrested L1s and seeded a 2.5 L flask with K media, 10X HB101, and the L1 populations. We grew this population shaking at 220 RPM at 20°C for 72 hours until gravid adults were present. We bleached the entire population to harvest embryos and allowed the embryos to hatch and L1s to arrest overnight in K media. These L1s were used to seed the RNAi experiment.

RNAi-sensitive pooled population RNAi experiment:

After we verified *pos-1* RNAi responses and harvested L1s from RNAi-sensitive individuals, we seeded experimental flasks. We performed two replicates for each condition. Each experiment contained 50 ml of K media, 4 ml of 10X bacteria (HT115 with an empty vector or an HT115 with gene targeting plasmid), and 75,000 L1s in a 250 ml baffled flask. We incubated the flasks in a shaking incubator at 220 RPM and at 20°C for 72 hrs. After 72 hrs we transferred the cultures to 50 ml conical tubes and allowed them to settle for 5 min, at which point we transferred 35 ml of the supernatant that contained F1 L1s. We used those L1s to seed a second generation of growth on RNAi in the same conditions described above. After another 72 hrs of growth, we separated F2 L1s from F1 adult pellets. We spun down the L1s at 1200 RPM for 2 min and saved the L1 pellet for sequencing. We also stored the F1 adult pellet for sequencing. DNA was extracted from these pellets and libraries were made as described below.

Sequencing

Genomic DNA was prepared using Purelink Genomic DNA Mini Kit (Invitrogen cat# K1820-01), libraries were constructed using the Nextera XT library kit (illumina cat# FC-131-1024), and sequenced on the NextSeq2000 platform.

Cross Construction

Before cross-construction, male stocks were generated by heat-shock-induced nondisjunction. For each strain, fifteen L4 hermaphrodites were transferred to three 6 cm NGMA plates and incubated at 30°C for 5 hours, and then returned to 20°C. After 72 hours, plates were screened for males and crossed to L4 hermaphrodites maintained at 20°C. For each male cross, 5-10 hermaphrodites and 10-20 males were placed on 6 cm NGMA plates that were seeded with 10 µL of OP50 and allowed to mate overnight. The following day, individual mated hermaphrodites

were transferred to fresh 6 cm NGMA plates, and the resulting progeny were monitored for an approximately 50:50 male-to-hermaphrodite ratio. Crosses for population construction were performed using the same mating procedure described above.

To control for cross-directional effects, reciprocal crosses were constructed for JU1793 x JU2466 and JU1793 x AB1. Each strain serves once as the hermaphrodite parent and once as the male parent. After mating, an equal number of plugged hermaphrodites were pooled within each reciprocal pair and plated onto 10 cm NGMA plates, yielding two independently amplified cross populations.

Cross Maintenance

Crosses were propagated for nine generations on 10 cm NGMA plates seeded with 1mL of OP50, with approximately 100,000 worms maintained per generation. When abundant L1 larvae were present, plates were washed with M9 buffer and gravity-settled to separate higher-density, older worms in the pellet from lower-density L1 larvae in the supernatant that would be seeded for the next generation. L1 larvae were titered prior to plating (~1,000 L1s per NGMA plate), and remaining larvae were intermittently frozen down for long-term storage.

For generation nine, the number of plated worms was doubled to ensure sufficient population size for three RNAi conditions and a time-zero sequencing control. The resulting F10 progeny were distributed across the three RNAi conditions and propagated using the same procedure, with remaining larvae from each generation saved for downstream sequencing.

NIL Construction

To validate the QTL on Chromosome III, multiple sets of Near-Isogenic Lines (NILs) were constructed and phenotyped. Heterozygous F1 animals were first generated by crossing JU1793 and JU2466. Approximately 72 hours later, F2 recombinant progeny were isolated by transferring single F2 animals into separate wells of a 96-well plate containing 50 uL of K media and 0.008 mg/ml HB101. After each well produced F3 progeny and reached starvation, 15 uL of the worm suspension was added to 15 uL of 2x lysis buffer containing 2% proteinase K. Samples were processed in a thermocycler at 60°C for 1 hour, followed by heat inactivation at 95°C for 20 minutes. After genotyping with primers targeting the regions of interest, starved F3 suspensions were recovered onto 6 cm NGMA plates seeded with OP50. Recovered F3 progeny were singled and re-genotyped to identify homozygous recombinant individuals. Confirmed NIL candidates were then backcrossed to their respective parental strain to reduce background variation further. Once the introgressed interval became too small for additional recombination, CRISPR-Cas9-mediated homologous recombination was used for further fine-mapping.

CRISPR Design to edit *sid-2*

We used the guide RNA gSZ182 to create a double stranded break near the position of the III:13680248 variant that differentiates JU1793 and JU2466 [JU1793:C>JU2466:A] at the SID-2 locus. Injection mixes were made with repair templates oZ446 or oZ456 to swap the JU1793 allele to the JU2466 allele and the JU2466 allele to the JU1793 allele, respectively. Injected animals were singled 16 hours after injection and allowed to generate progeny. Roller F1 progeny were singled and allowed to lay progeny for 24 hrs at which point the F1 roller parent was lysed for DNA extraction. The primers oZ461 and oZ468 were used to amplify the region surrounding the edit site. PCR amplicons (10 µl) were subsequently digested with HpyCH4IV using the cutsmart buffer.

Phenotyping

Strains were synchronized by hypochlorite treatment and grown until L4s were visible. Three days prior to plating onto RNAi bacteria, populations were amplified by transferring five L4s to each of three plates. After approximately 72 hours, 10-15 L4s were plated onto each RNAi condition. The following day, 150 and 200 embryos were transferred to an unseeded NGMA plate and counted to determine the exact starting number of embryos. After 24 hours, the number of unhatched embryos was counted. RNAi sensitivity was quantified as the fraction of embryos remaining unhatched divided by the total number of embryos plated.

Make oligo table + guide RNAs

Organized the following reagent tables. They are currently in the SZ stocks google document:

Strains

Oligos

gRNA

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